

PVEMON

PVE nodes and backups monitor

User guide

Version 1.5.2

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MIT licence · Dagarin

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1. What it is and what you can do

PVEMon is a web application that shows you, at a glance, whether the Proxmox servers, virtual machines, LXC containers, Docker containers and backups assigned to you are working properly, and warns you when something fails. This guide explains how to use it step by step; it contains no keys or internal addresses.

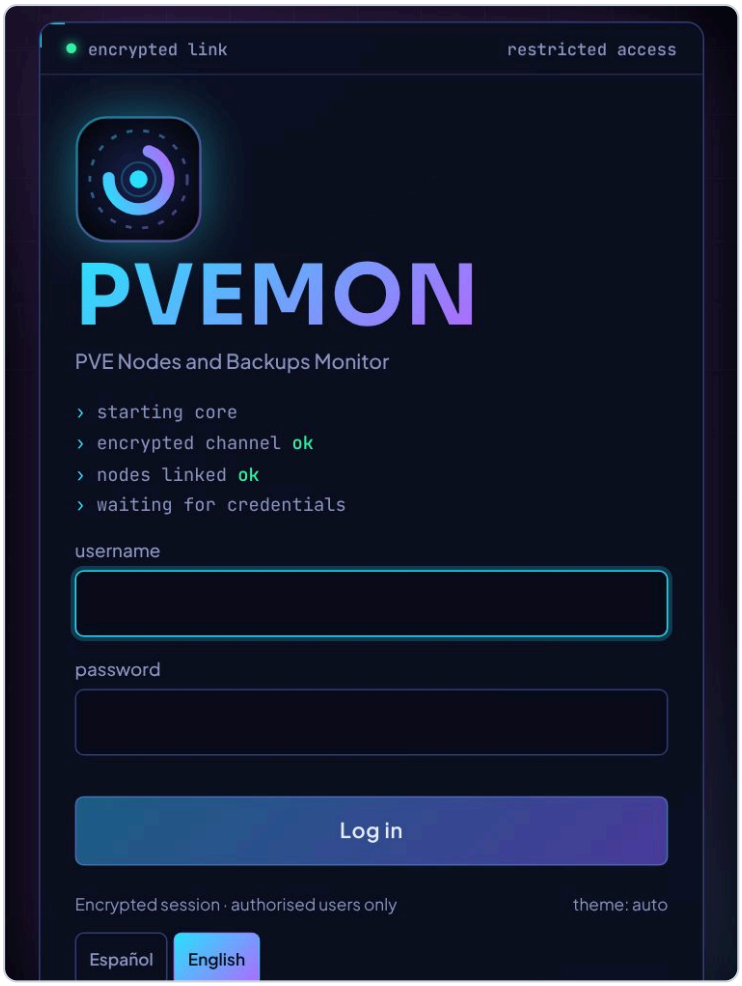
With your user you can:

- See the status of **only** the items the administrator has given you.
- Look at usage charts (CPU, memory, disk, network).
- See the alert history and the status of your backups.
- Choose how you want to be notified: phone notification, email or Telegram.
- Install PVEMon as an app on your phone or computer.
- Use it in **Spanish or English**, whichever you prefer.

> PVEMon **does not allow** switching off, switching on, deleting or changing anything: a *monitor* user is read-only and only sees what the administrator has marked as “Visible” for them.

2. Logging in

Open the address the administrator gave you, type your username and password and press **Log in**. The welcome screen shows the PVEMon icon above the name. If your account has two-step verification (2FA), it will also ask for the 6-digit code from your authenticator app. Below the form there is an **Español / English** selector: Spanish is the default language.

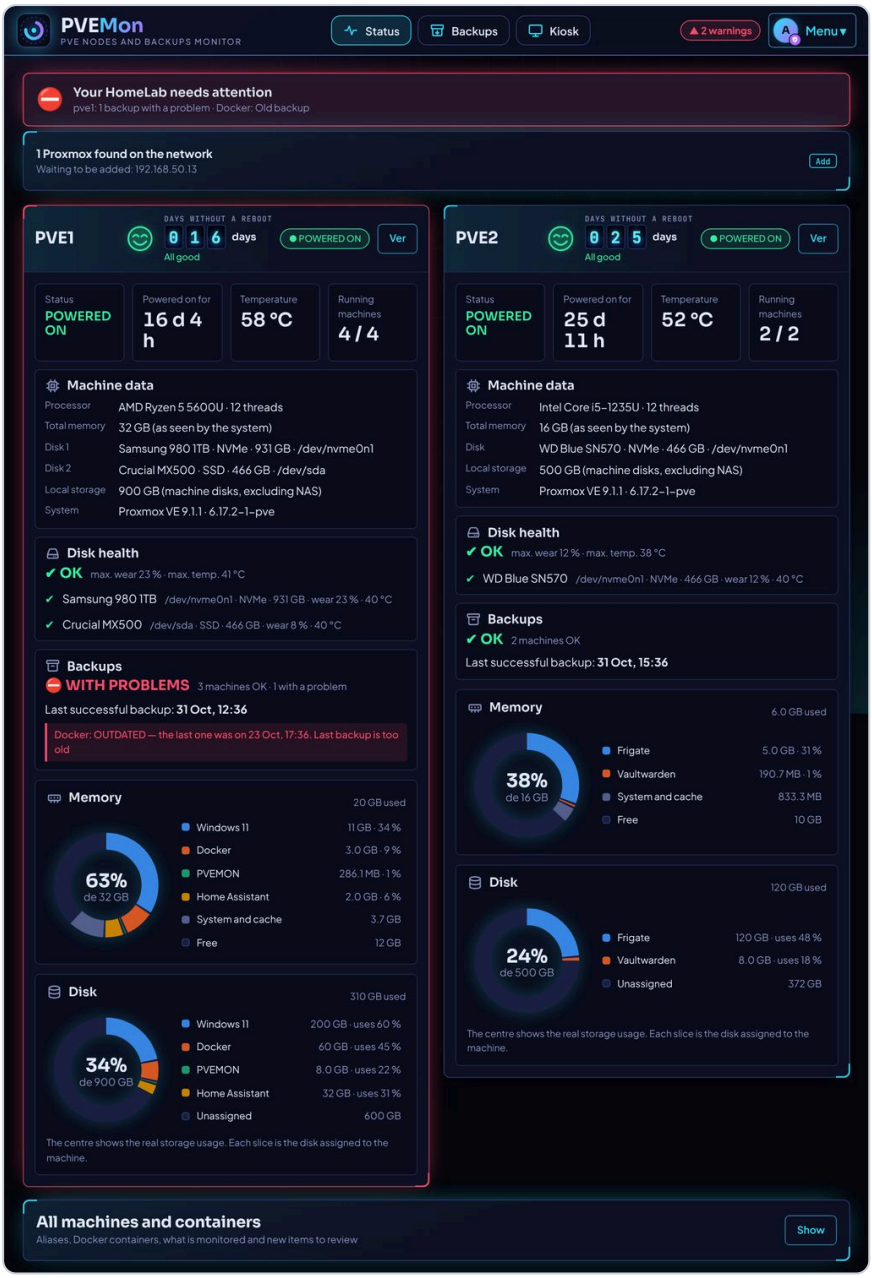


Login screen

- After several failed attempts the account is locked for a few minutes: wait and try again.
- The session closes by itself after a long time; just log in again.
- Always log out (menu, “Log out”) if you use a shared computer.

3. The Dashboard

This is the home screen (the **Status** button). It shows one card per node with its state, uptime, temperature, running machines and, next to the name, the **uptime counter** (days without a reboot, with a little face that changes mood). Below it there are four blocks in the same style: **Machine data**, **Disk health**, **Backups** and the **Memory** and **Disk** rings.



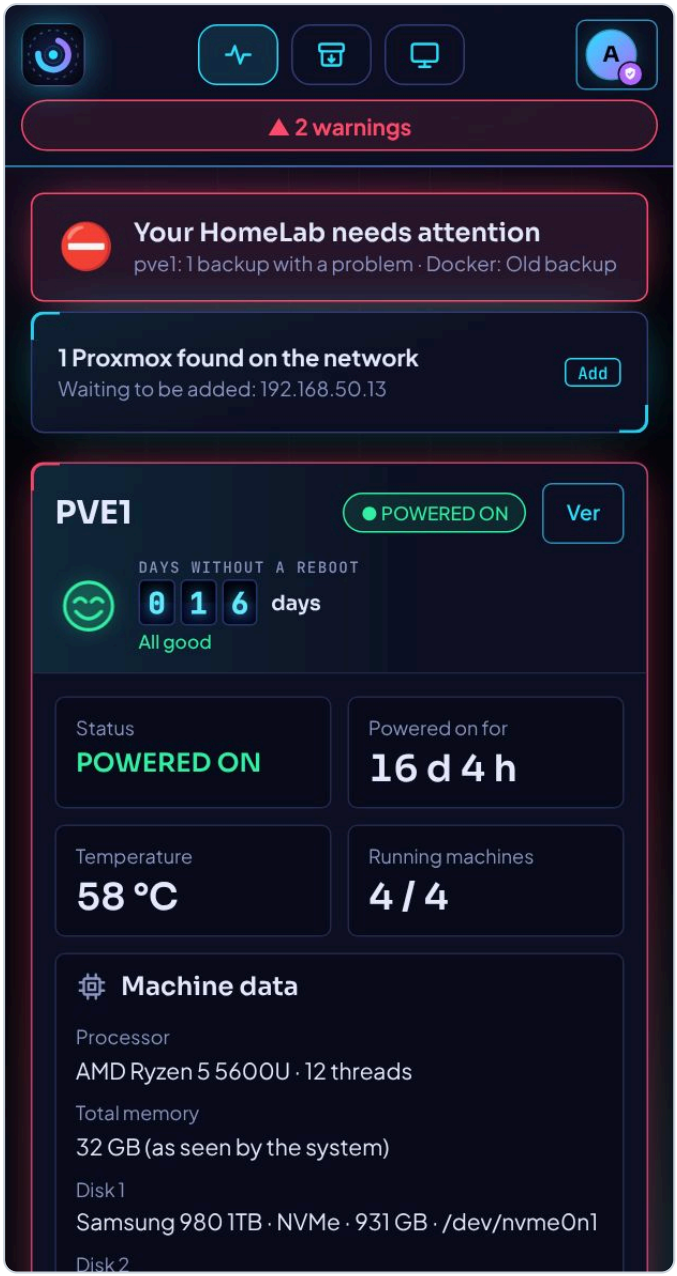
Main dashboard on a computer

- **Machine data** identifies the processor, the memory and **all the physical disks** of the node (model, type, size and device), not just the system one.
- **Disk health** summarises the SMART/ZFS state, the maximum wear and temperature, and lists each disk with a ✔ or ⛔ mark.
- **Backups** tells you whether the latest backup of each machine is good, when it was made and which ones have problems.
- In the **Memory** and **Disk** rings each slice is a machine; the centre is the real usage.

The colours mean: **green** everything is fine, **amber** warning and **red** down or with an active alert.

> There is no “all good” message any more: when everything is fine the screen stays calm, and the red banner (“Your HomeLab needs attention”) and the alerts pill only appear when

something fails. Pressing them opens the **first item with a problem** directly.



Dashboard on a phone

The data refreshes by itself while the screen is open (it pauses in the background to save battery). The **Show** button of "All machines and containers" unfolds the full list.

4. Title bar and menu

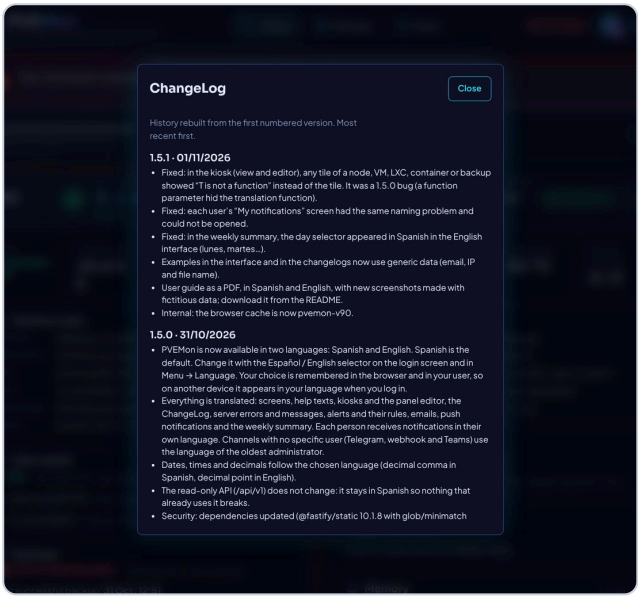
On a computer, the bar has the logo and name on the left, the **Status**, **Backups** and **Kiosk** buttons in the middle and, on the right, the active-alerts pill and the menu. On a phone the first line holds the logo, the three icons and your photo; if there are alerts, they appear on a **line of their own** in red so they stand out.

- **Status**: the main dashboard. **Backups**: the backup summary.
- **Kiosk**: unfolds a menu to choose which kiosk to open full screen; the one marked ★ Main comes first (section 9).

The menu gives access to Dashboard, Alerts, Backups and Customise (your profile), and lets you choose the theme (Auto, Light or Dark), the **language** (Español or English) and **Log out**. Administrators also see Nodes, VMs, LXCs, Docker, Kiosks and Settings. At the bottom you will find the version, the **ChangeLog** button and the "© Dagarin 2026" notice.



Menu on a phone (administrator)



ChangeLog window

The **ChangeLog** is the history of what is new in each version, from newest to oldest.

5. Machine details

Pressing a machine shows its sheet: state, uptime, backup data, memory and disk rings (used versus free) and the CPU, memory, disk and network charts. For VMs and LXCs you also see its **IP address** and, for VMs, whether the guest agent (qemu-guest-agent) is responding. You can change the chart period (1 h, 6 h, 24 h, 7 d or 30 d) and hover or swipe over a chart to see the exact value.



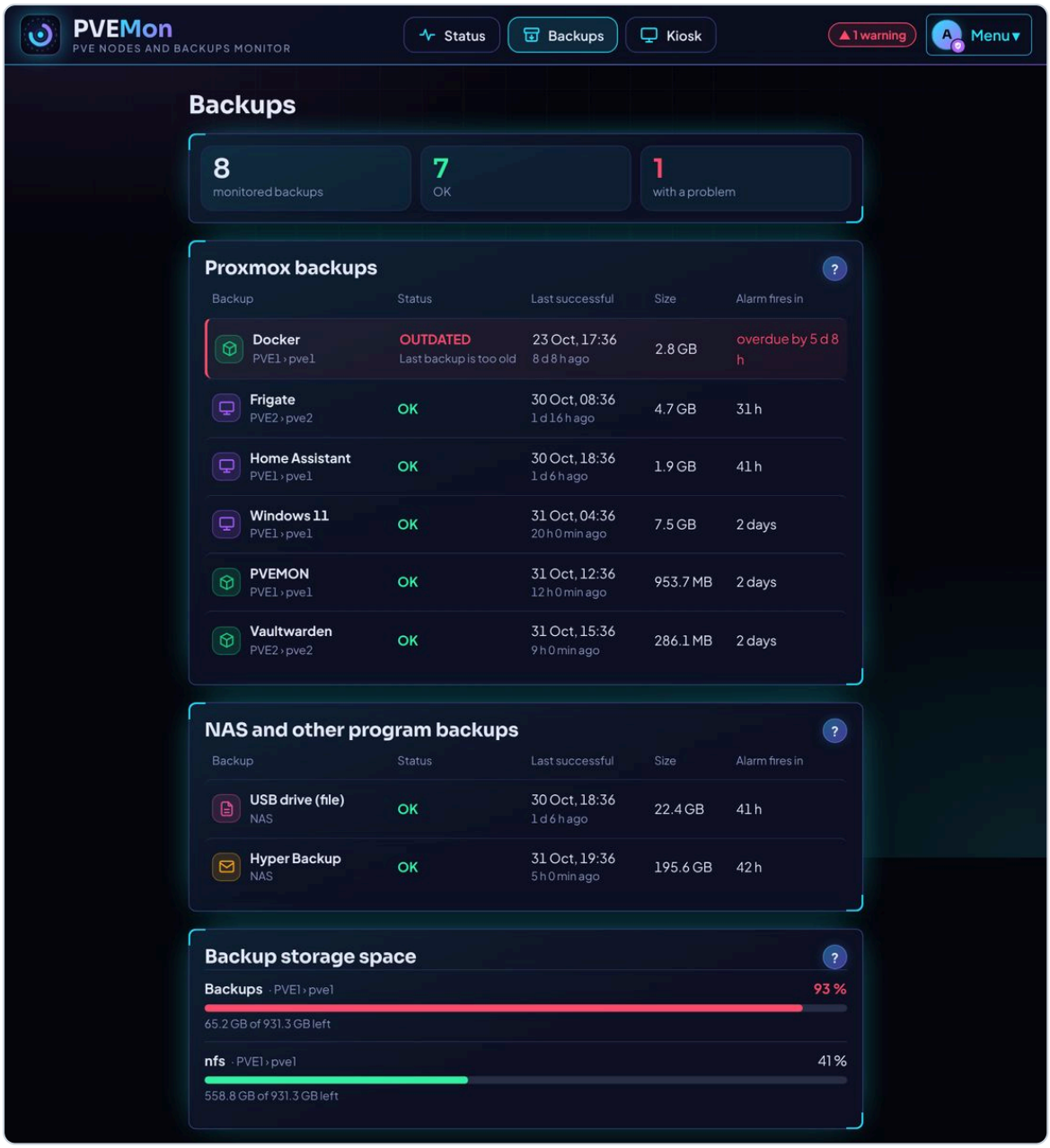
Sheet of a virtual machine (top part)

The **?** button next to many titles opens a short help text explaining that section.

6. The Backups screen

Backups (title-bar button or menu option) summarises all the backups assigned to you: Proxmox ones, those from a NAS or other programs that report by email, and file checks on a network disk. Each backup shows the date of the latest good one, its size and the **time left before the alarm** if a new one does not arrive; whatever fails appears at the top.

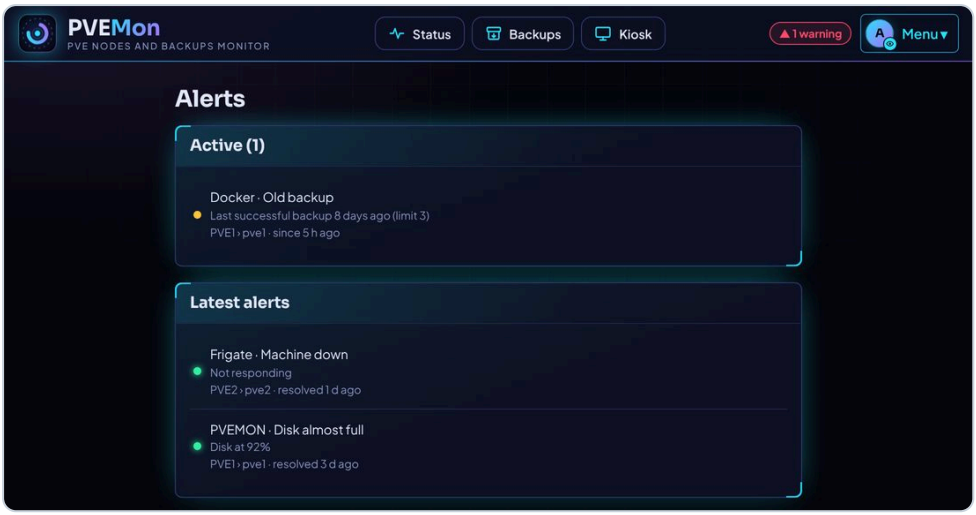
An icon tells the backup type apart: Proxmox container or machine, an **envelope** for backups reported by email and a **document** for file checks on a network disk. At the end it shows how full the backup storages are: amber from 80 % and red from 90 %.



Backup summary

7. Alerts

The **Alerts** screen shows the active alerts and the latest resolved ones for the items you can see: when they fired, which item was affected and how long they have been open. An alert fires, for example, when a machine goes down, the CPU or disk fills up, a backup is old, a disk fails or wears out, or a Docker container stops.



Alerts

Administrators can **snooze** an active alert (for example 1 hour) from Settings → Alerts: while it is snoozed it does not count as active and does not notify. If the problem is still there when the time is up, you are notified again. In the snoozed list there is a **Reactivate** button.

You cannot change the limits: the administrator sets them. What you do decide is **how you find out** (section 10).

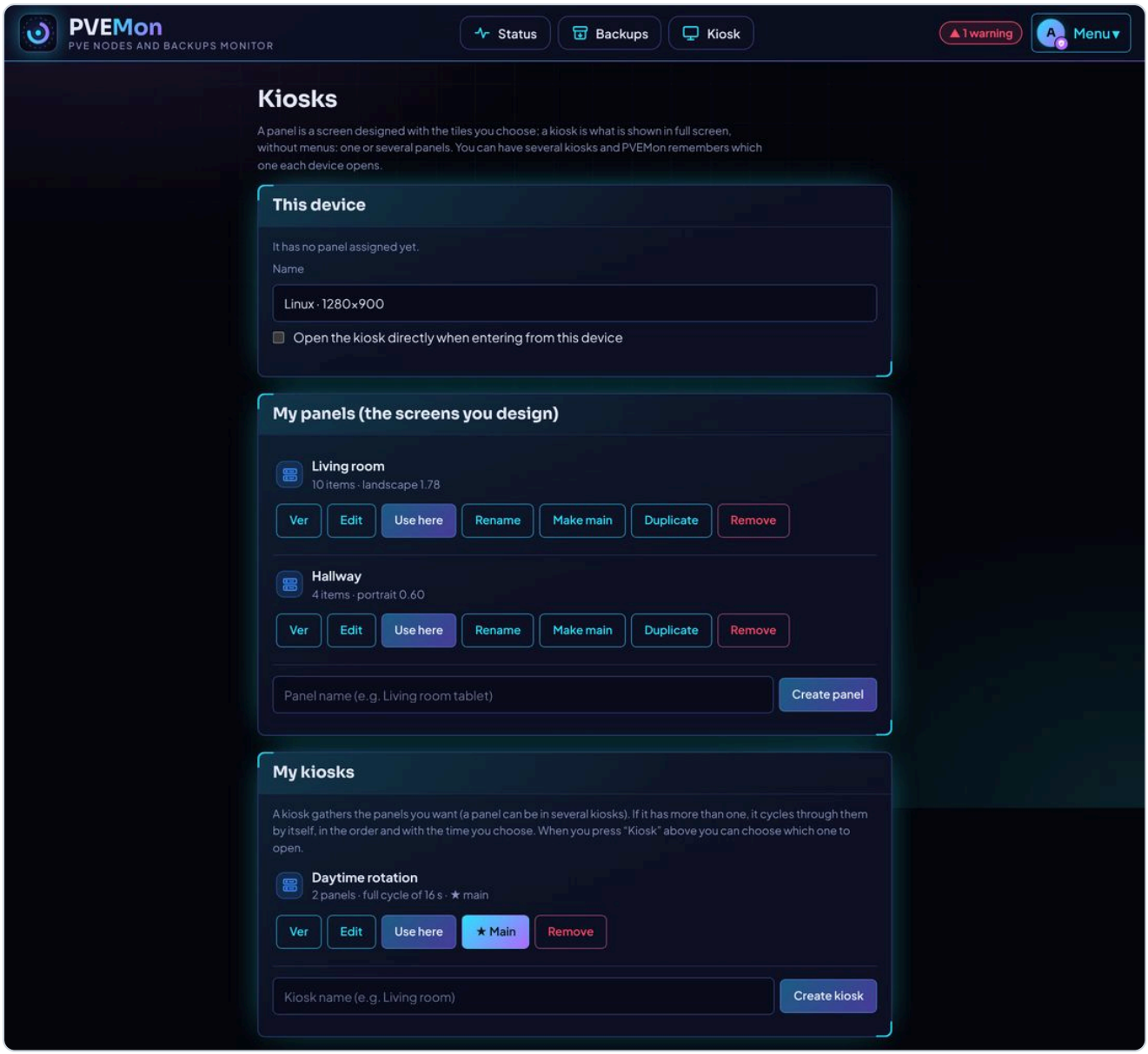
8. Disk health

PVEMon reads the state of the disks of every monitored node every 30 minutes: SMART health, SSD **wear**, **temperature**, power-on hours and **ZFS** pools. On a node sheet, the **Disks** card (just before “Connection and maintenance”) lists each disk; on the Dashboard it is summarised in “Disk health”.

- A disk with a failed SMART test, pending sectors or media errors is marked ⛔ and the “Failing disk” alert fires.
- “High SSD wear” fires above 90 % and “Disk running very hot” above 65 °C (the administrator can change both).
- If a node has several disks, all of them appear, each with its model and device.

9. Kiosk

A **kiosk** is a clean screen, with no bars or menus, meant for a tablet or an always-on monitor. Each user can have several **panels** (each with the tiles they choose) and group them into **kiosks**, which cycle from one panel to the next. They are managed in **Menu → Kiosks**.



Kiosks: this device, my panels and my kiosks

Panels and tiles

- **Canvas editor**: **+ Item** adds nodes, VMs, LXCs, Docker containers and backups; also **summary** tiles (Systems status, Backups status) and **text boxes** in six font sizes.
- Each tile shows its name, an LED (green running, red stopped) and, if you wish, CPU, memory and disk as needle gauges.
- They move by dragging, stretch from any side or the corner, and the **MINI** button leaves only icon, name and LED.
- To move several at once, draw a box around them (or Shift/Ctrl + click).
- **Cmd/Ctrl + Z** undoes the last change (repeatedly) and **Cmd/Ctrl + Shift + Z** redoes it; the **arrow keys** move the selected tile (faster with Shift).
- The screen shape is chosen in **Screen** (16:9, 4:3, portrait, this device's, or **Custom…** with width × height) and each panel has its own **theme** (Auto, Light or Dark).



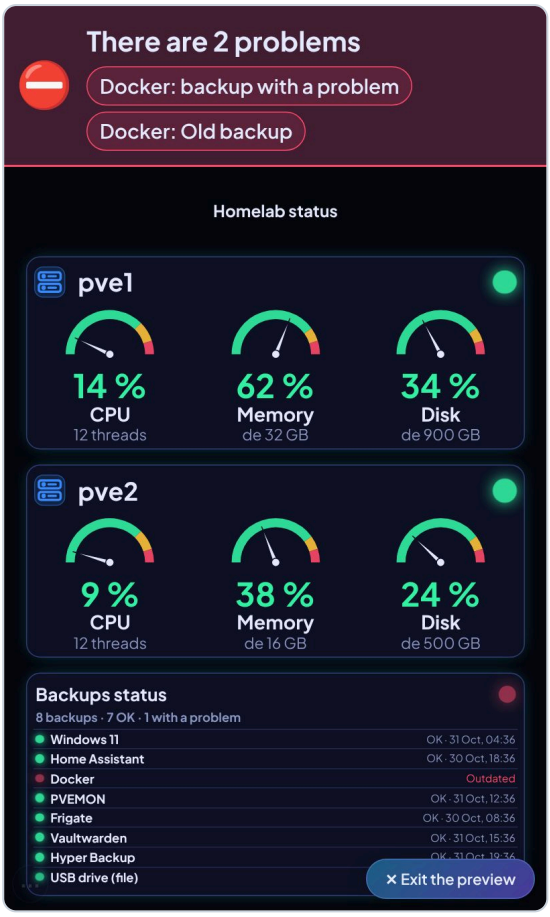
Panel editor

Viewing the kiosk

If there is any problem, an **error band** appears at the top with a button for each warning; pressing it opens the affected item. Pressing a tile opens its item and the back arrow returns to the kiosk. The screen stays on and refreshes itself every 10 seconds. The ⋯ button gives access to full screen, “Switch kiosk”, My panels and exit.



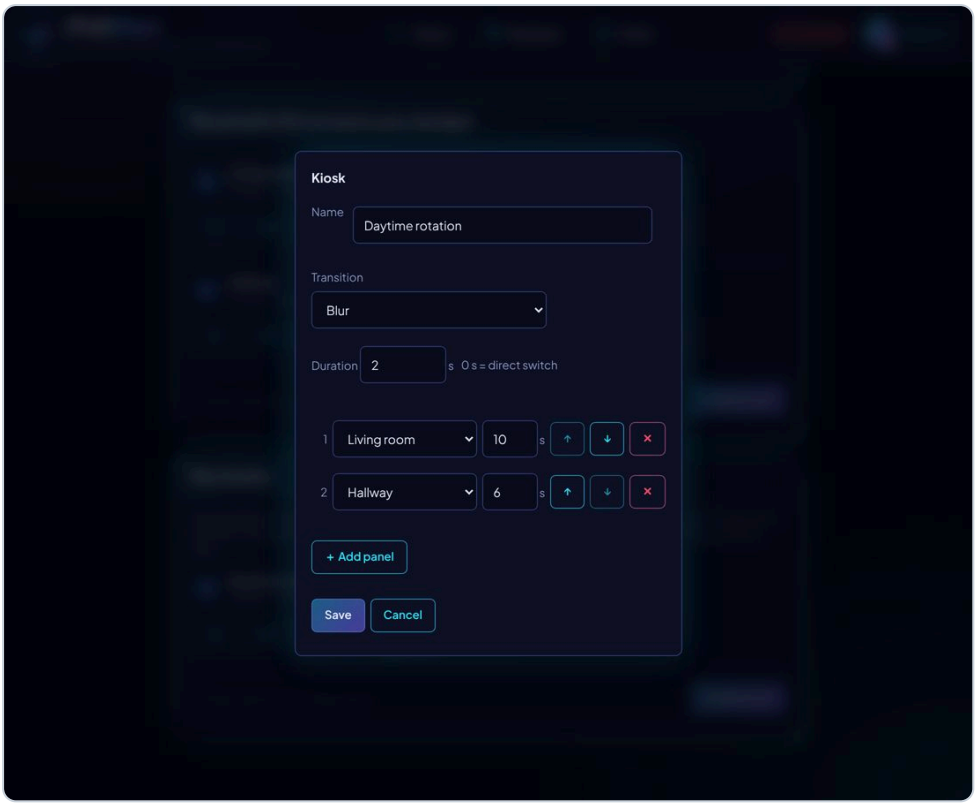
Landscape panel (Living room)



Portrait panel (Hallway)

Kiosks, main kiosk and devices

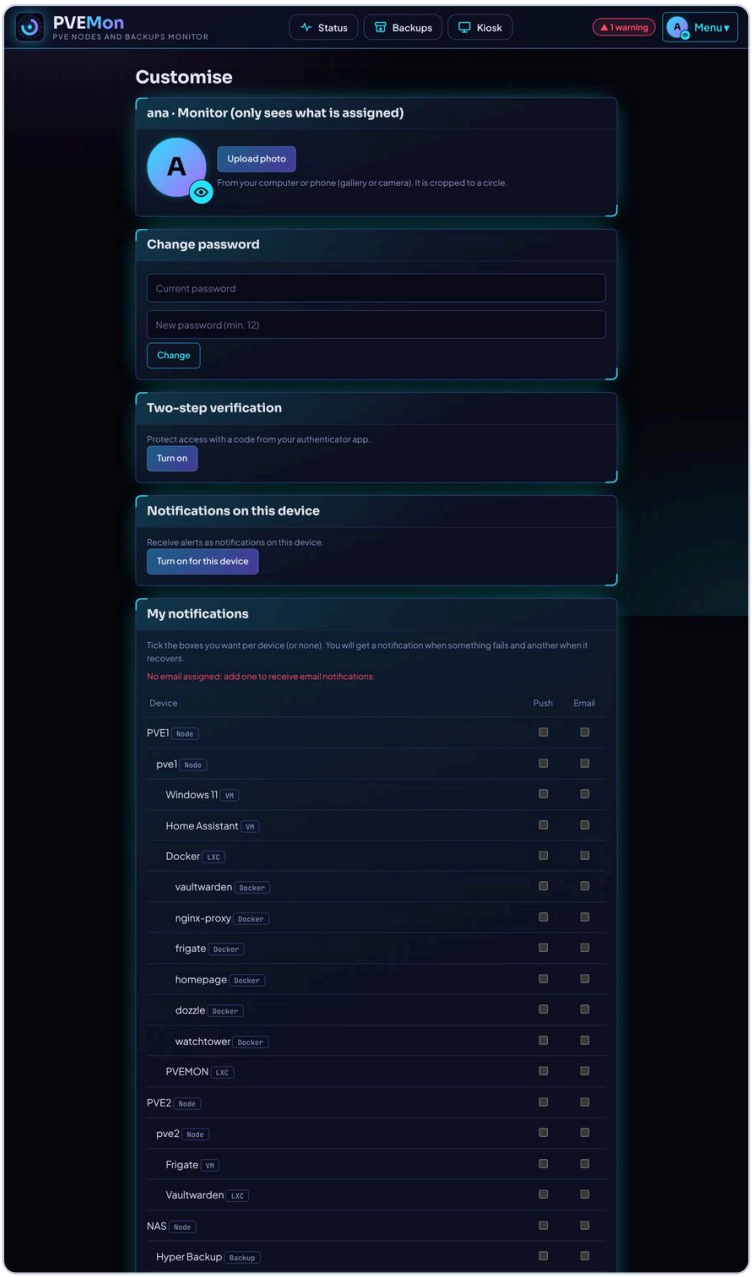
- A **kiosk** gathers the panels you want (the same panel can be in several) and cycles through them in the order and for the time you choose. The **transition** can be **pixel dissolve** or **blur**, with configurable duration (0 s = direct switch).
- The ★ **Main** button marks the panel or kiosk that comes first in the menu of the **Kiosk** bar button.
- **Use here** assigns a panel or kiosk to this device (each browser keeps its own code); with the “Open the kiosk directly when entering” box ticked, the device starts straight in kiosk mode.
- On a **phone**, inside a kiosk, swipe left or right to go to the next or previous panel.



Kiosk editor

10. Customising your account

In the menu, go to **Customise** to change your profile **photo**, your **password** (use a long one that you do not use elsewhere) and to turn on **two-step verification (2FA)** with an authenticator app, which is highly recommended. You can also enable notifications on this device and choose your notifications. The **language** (Español or English) is changed in **Menu → Language**; it is remembered on your user and is also used in the emails, push notifications, Telegram messages and summaries you receive.



Customise and My notifications

My notifications

For each item you can see, you can tick which channels should notify you when there is a problem and when it recovers:

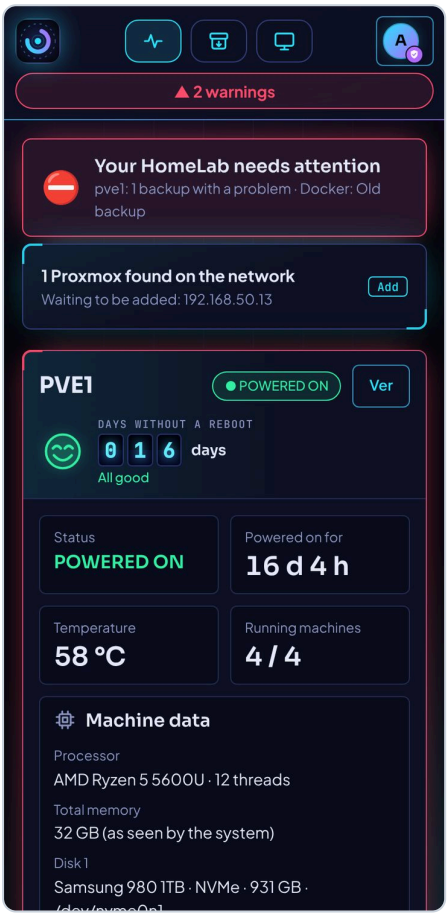
Channel	What you need
Push	Install PVEMon as an app and accept notifications on that device
Email	The administrator must have email configured; it arrives at your account's address
Telegram	The administrator must have enabled Telegram and told you how to link it

You can only choose notifications for items assigned to you.

11. Installing PVEMon as an app

PVEMon can be installed without an app store:

1. **iPhone/iPad (Safari)**: Share button → **Add to Home Screen**.
2. **Android (Chrome)**: ⋮ menu → **Install app**.
3. **Computer (Edge/Chrome)**: install icon in the address bar.



On a phone: Dashboard



On a phone: Nodes (administrator)

If something looks odd after an update, hard-reload the page (Cmd+Shift+R on Mac, Ctrl+F5 on Windows).

12. Privacy and access limits

- You only see the items marked as **Visible** for you; the rest do not appear in lists, alerts or charts.
- You cannot enter the administration screens (nodes, settings, users) or change anything on the servers.
- PVEMon does not show you passwords, keys or internal server addresses.

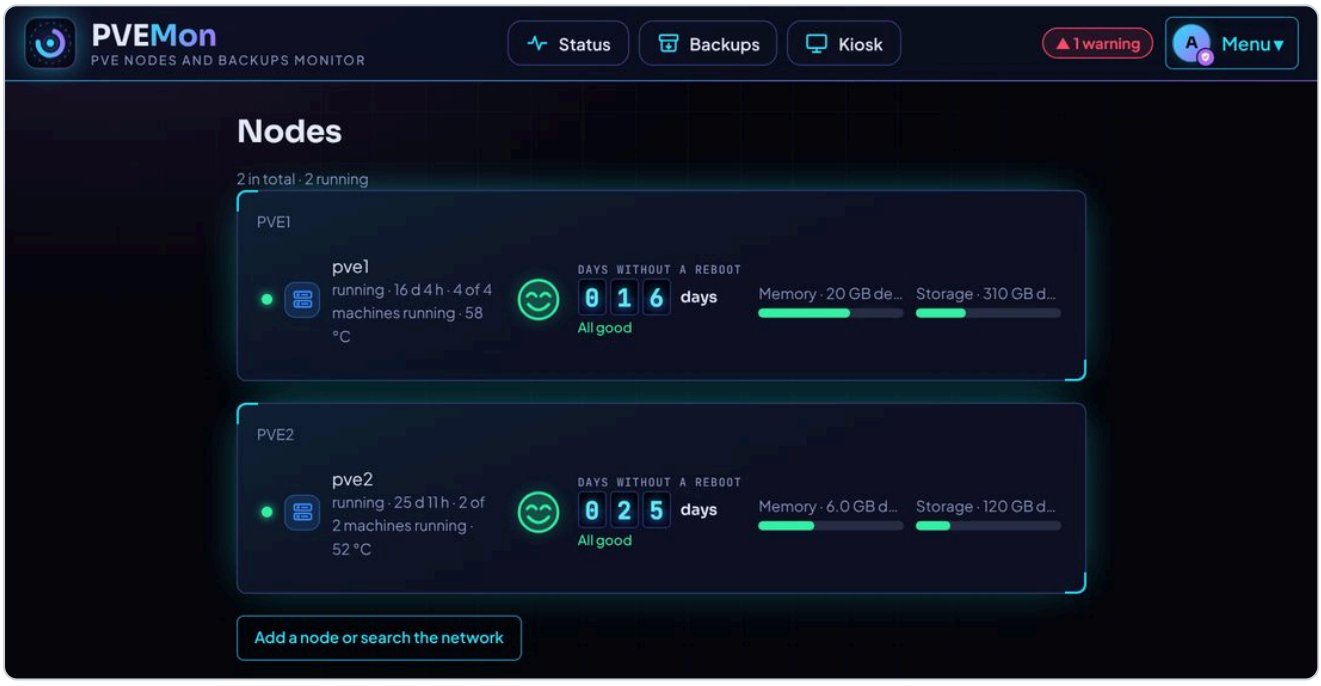
- Important actions are recorded in an audit log for the administrator.
- If you suspect someone knows your password, change it and tell the administrator.

Administrator appendix

This part is only useful to whoever administers PVEMon; a user with the monitor role does not see these screens.

A1. Nodes: list, sheet and adding

Nodes is a list with the state, uptime, running machines, temperature and memory and storage bars of each node. Pressing one opens its **sheet**.

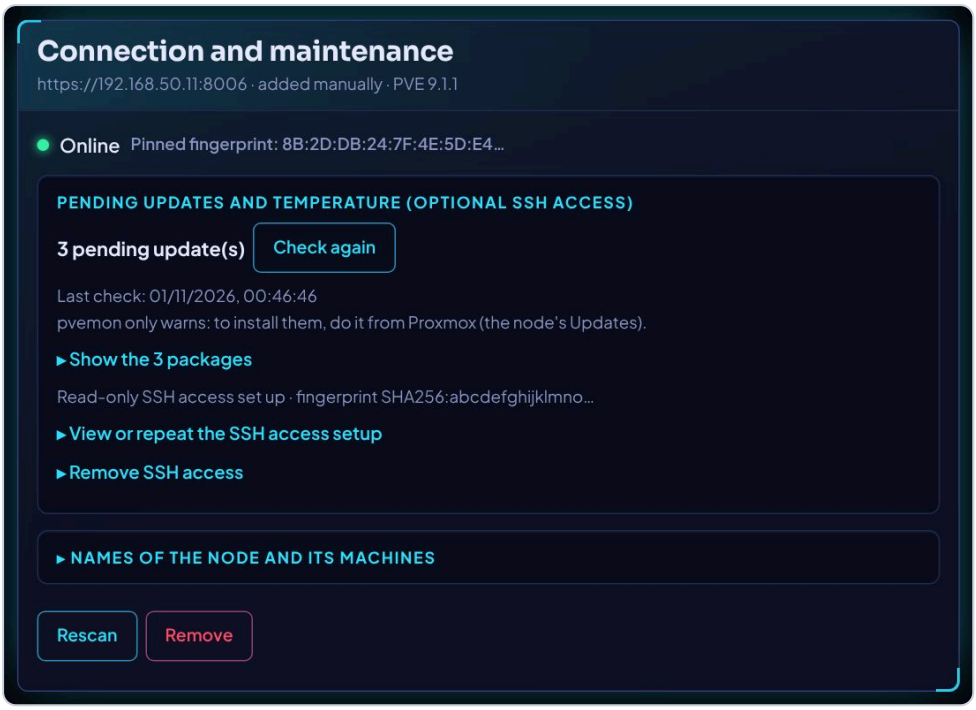


Node list

The sheet gathers the machine data (processor, graphics card, memory, **all the disks**, Proxmox version, kernel, the address PVEMon uses to connect), the memory and storage rings with used and free amounts and, in this order, the **Disks** card (health, wear and temperature of each), the **Connection and maintenance** section and the charts.

Connection and maintenance shows the connection state and the pinned server fingerprint, the **pending updates and temperature** section (optional SSH access), the names of the node and its machines, and the Rescan and Remove buttons.

> **PVEMon only reads.** It does not install updates, reboot or change the node: it only **tells you** how many are pending and which ones; to install them, do it from Proxmox. Without SSH it uses the data Proxmox already has (it can take up to a day to reflect new things).

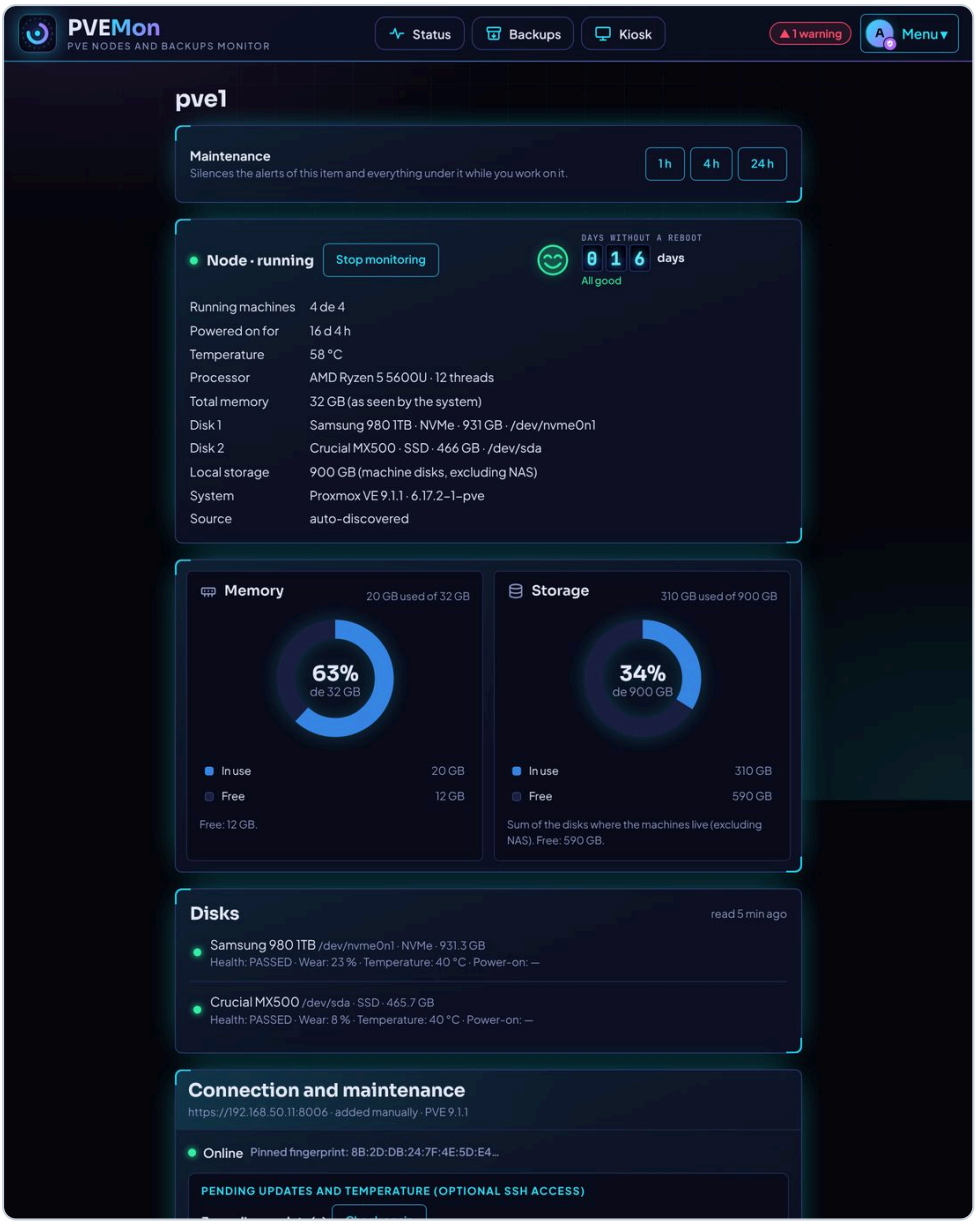


Pending updates and optional SSH access (read-only)

Optional SSH access (read-only)

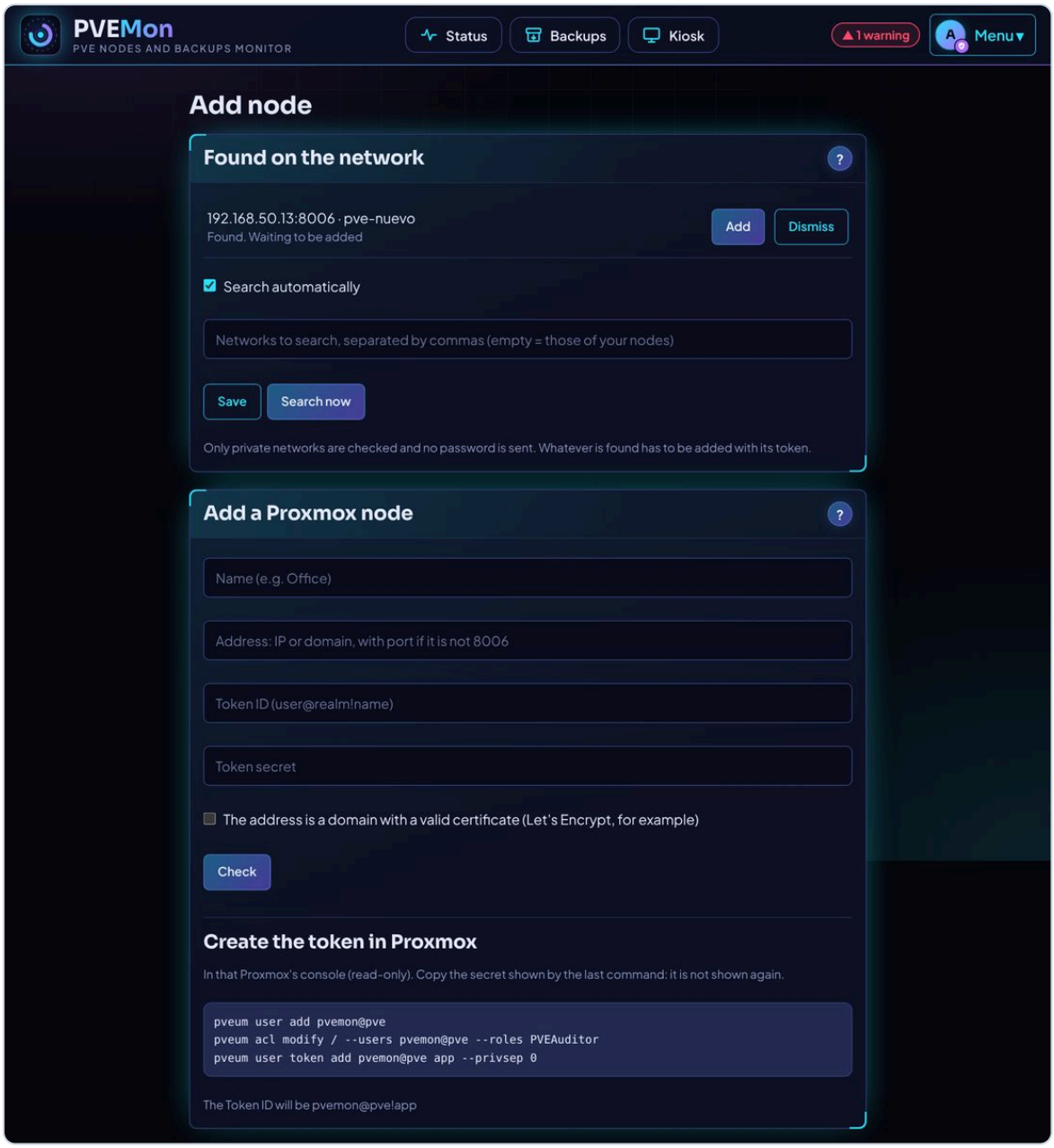
If you paste the “Prepare SSH access” block once into the node’s console, PVEMon can check pending updates at once and read the CPU temperature. The block creates a PVEMon-only key restricted (`restrict` + fixed command) to a script with **three read-only commands**: `check` (lists packages not yet installed), `temp` and `version`. If you had pasted an earlier version, paste the block again: it overwrites the script and removes the write commands.

The **Remove SSH access** section shows the block that removes the key and the script from the node (and leftovers of old versions); then press **Forget access** so PVEMon stops using it. Without SSH access you lose the CPU temperature reading and the immediate check.



Node sheet (top part): data, memory, disks and connection

The **Add a node or search the network** button at the end of the list opens the add screen. PVEMon looks for Proxmox on your network and offers them to add; you can also add one by hand with its address and a **read-only** API token created in Proxmox (the screen shows the commands to create it). PVEMon never needs write permissions on the server.



Add node

A2. VMs, LXCs and Docker

VM and **LXC** are identical lists: the machines grouped by node, with their state, uptime and memory and disk bars (used over assigned; on VMs, the disk is read from the guest agent if active). Pressing one opens its sheet with memory and disk rings (used and free) and charts.

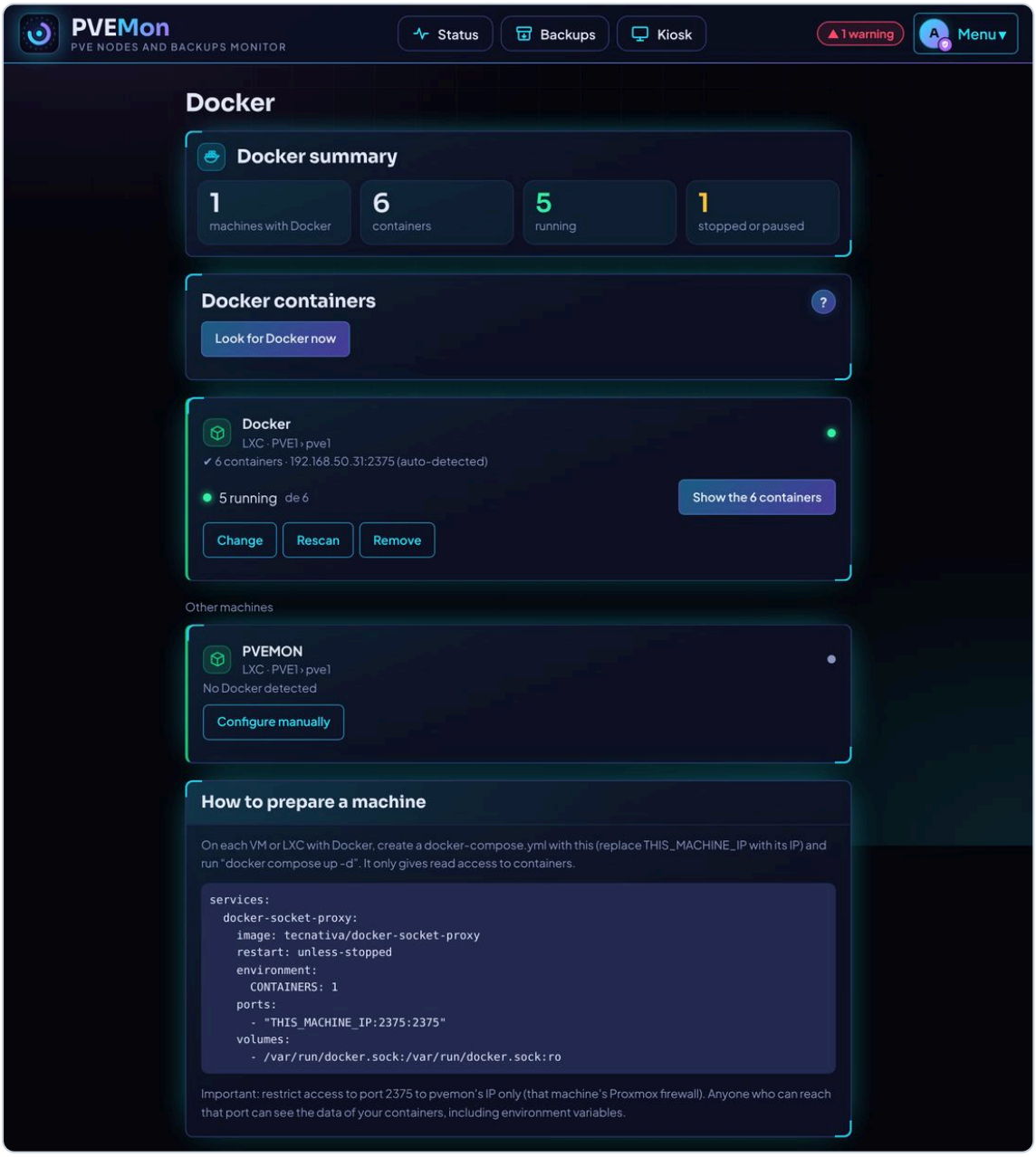


VM list



LXC list

Docker summarises the containers of each machine running Docker; press **Show the N containers** to unfold the list with their state and image. The “not monitored” and “new” notices help you decide which containers to watch.



Docker

Maintenance mode

On the sheet of a node, VM or LXC, the **1 h**, **4 h** and **24 h** buttons silence the alerts of that item and of what hangs from it while you work on it. A “maintenance” label appears and you can end it early. Sheets do not refresh by themselves while you have a window or dropdown open or are typing.



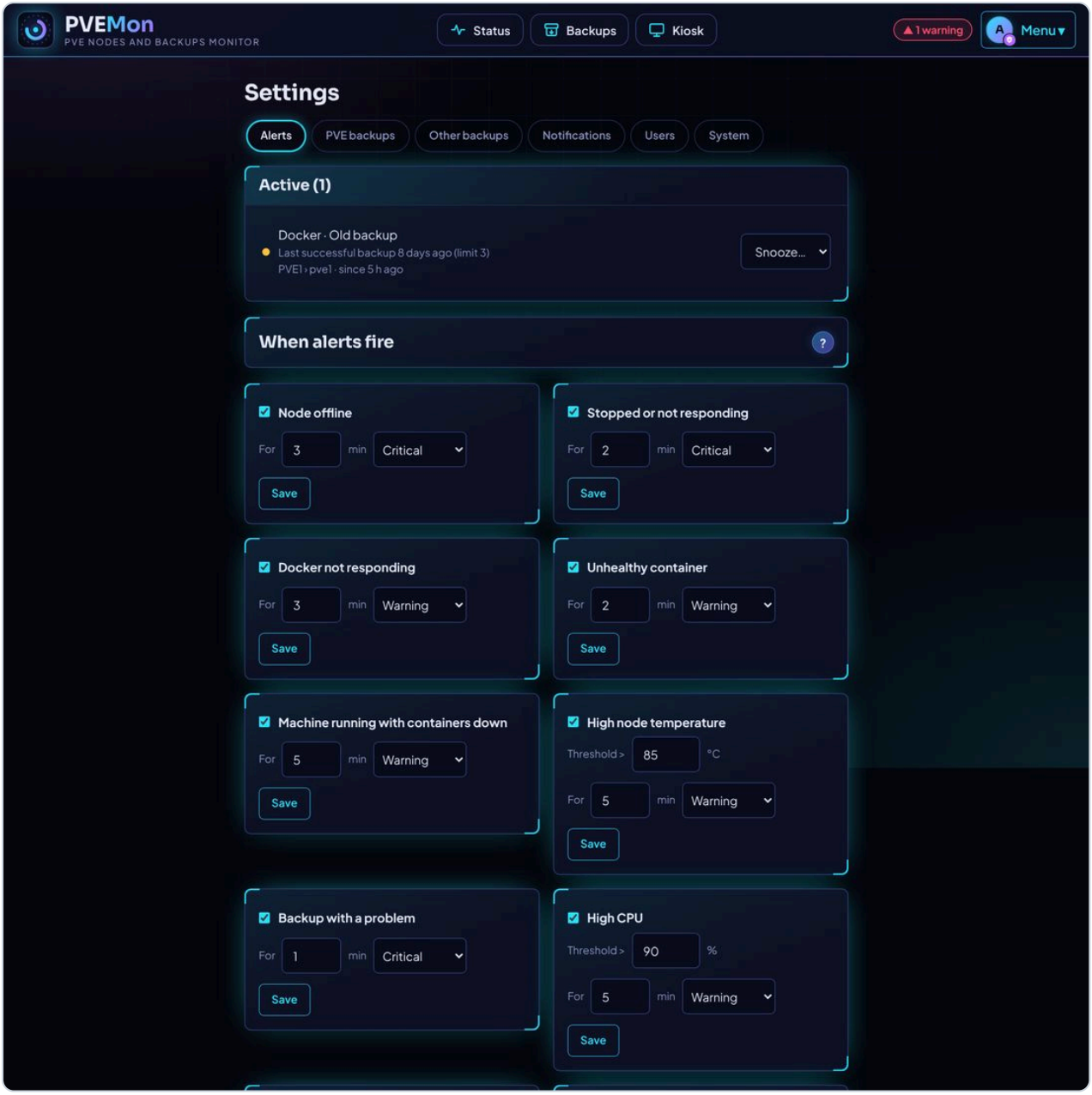
Sheet of an LXC (top part)

A3. Settings

Settings is split into tabs of equal width:

Tab	What it is for
Alerts	Active and snoozed alerts (with Snooze/Reactivate button), when they fire (threshold, duration and severity) and the history of the latest ones
PVE backups	Proxmox backups: how often a good backup must arrive, per server
Other backups	Backups from a NAS or other programs (by email) and file checks on a network disk (NFS)
Notifications	Email (SMTP), Telegram, generic webhook and Microsoft Teams
Users	Sign-ups, roles, password, two-step verification (2FA) and access to each item
System	PVEMon backup, external heartbeat, weekly summary and system data

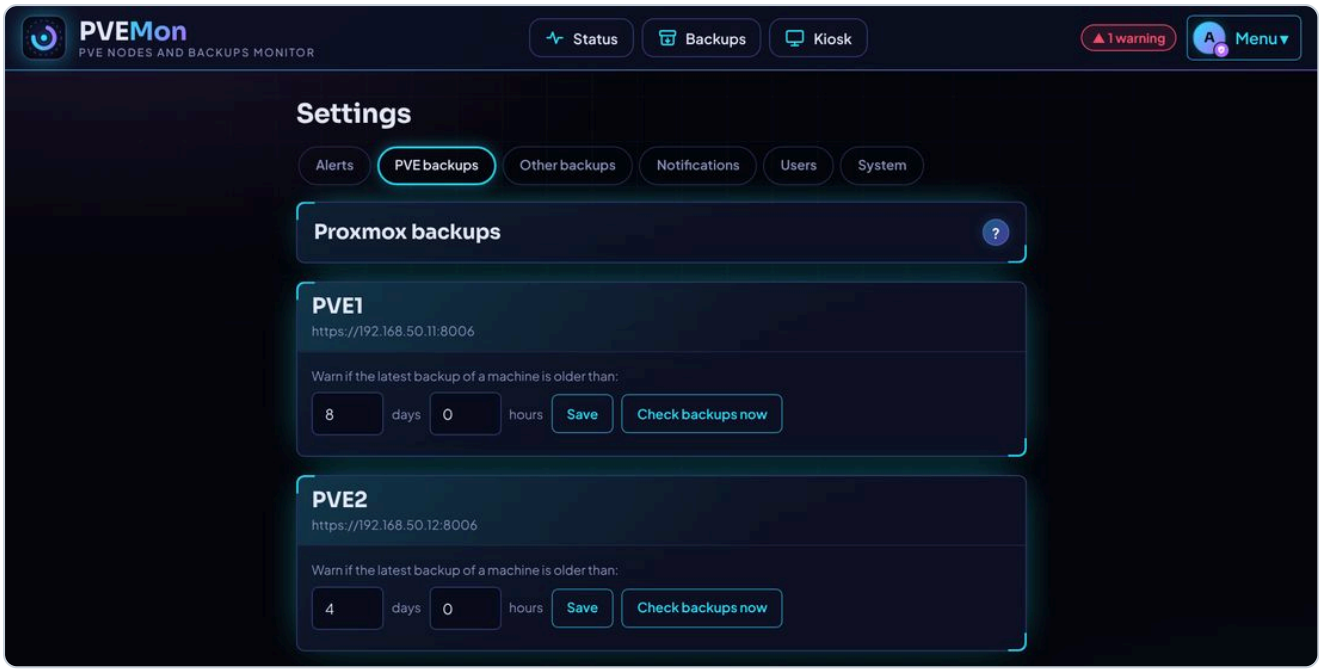
Alerts



Settings: alerts

PVE backups

For each server you set how many days and hours can pass without a good backup before the alarm fires. The **Check backups now** button forces a review. PVEMon reads the backup size from the Proxmox task log, without needing write permissions.



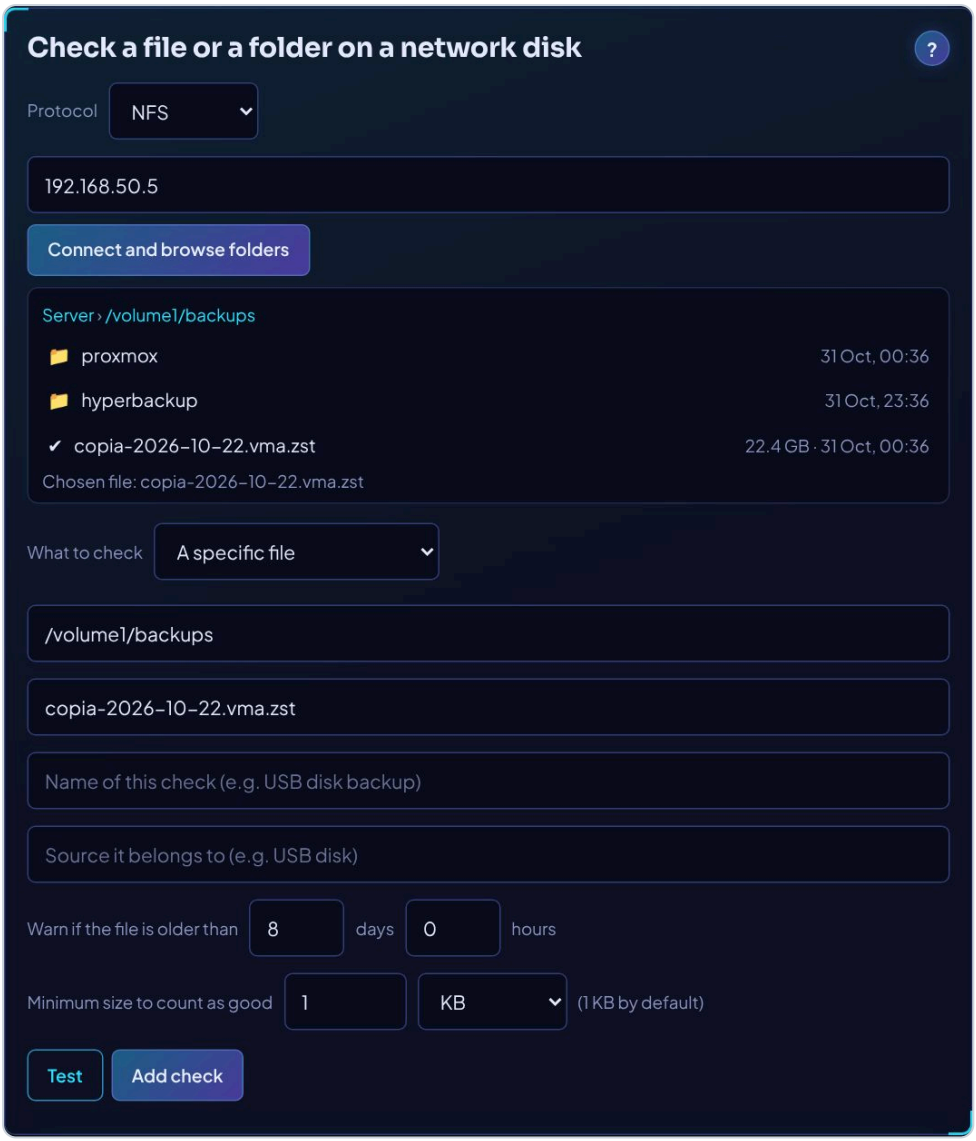
Settings: Proxmox backups

Other backups: backups by email and file checks

For programs that only report by email (Synology Hyper Backup, Cobian...) you configure the IMAP mailbox, the source of each backup and how many days can pass before a good one must arrive; the total size of the backup is read too. Senders already handled elsewhere can be excluded, and emails PVEMon does not understand are listed separately.

For those that do not report, the **Check a file or a folder on a network disk** card looks directly at the NAS:

1. Choose the protocol: **NFS** (SMB and FTP appear as “in development”).
2. Type the server and press **Connect and browse folders**: the exported folders are shown and you can browse them without typing paths.
3. Choose **a specific file** or **the newest file in a folder** (picked at each check by modification date, without subfolders or hidden files).
4. Set the **maximum age** (8 days by default) and the **minimum size** (1 KB by default; for example, 500 MB). If the file does not exist, is smaller or older, the backup becomes “WITH ERROR” or “OUTDATED” and the alert fires.



Check a file on a network disk

It is checked every 5 minutes (there is a **Check now** button) and uses the same notifications as the rest of the backups. PVEMon speaks NFS version 3 directly and only reads size and date: it mounts nothing and needs no privileges. On the NAS you must export the folder over NFS (with NFSv3 enabled) and allow PVEMon's IP.

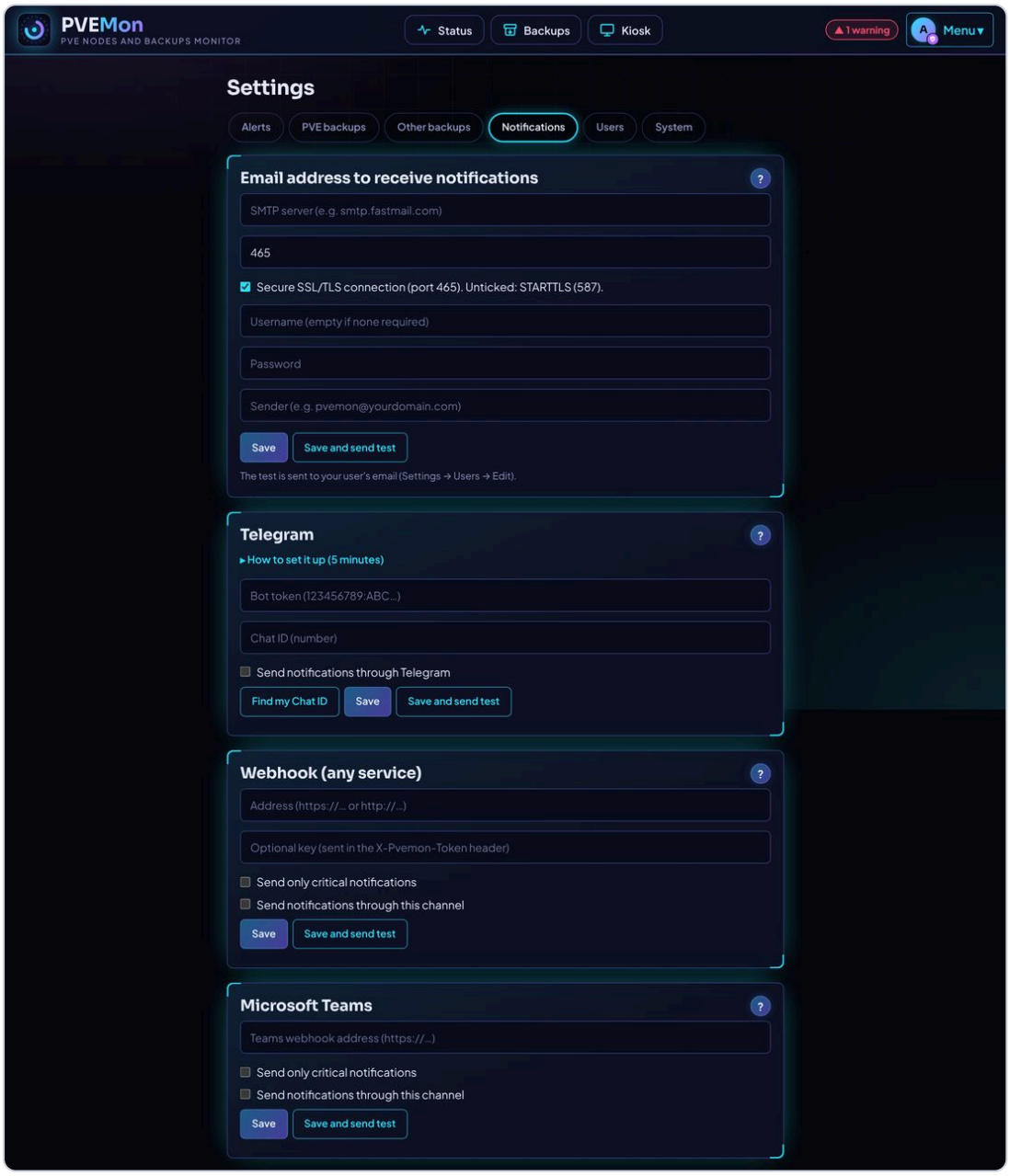
Snoozing alerts

In Settings → Alerts, each active alert has a **Snooze…** dropdown (for example 1 hour). It moves to the **Snoozed** list, does not notify or count as active and, if it is still failing when the time is up, you are notified again. **Reactivate** puts it back to active at once. The default rules now include "Failing disk", "High SSD wear" and "Disk running very hot".

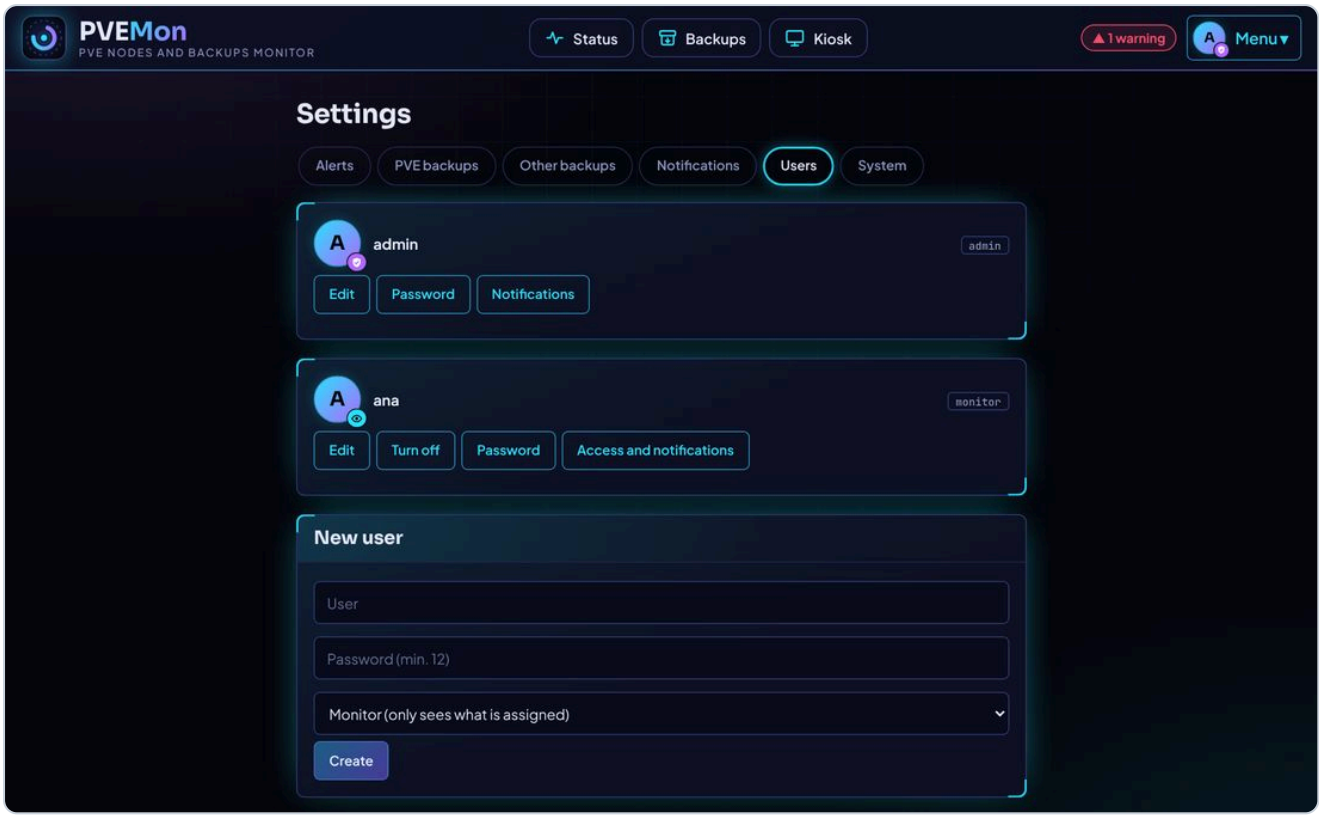
Notifications and users

In **Notifications** you configure the mail server (SMTP), the Telegram bot, a **generic webhook** (PVEMon sends a JSON with the alert to the address you give, with an optional token in the `X-Pvemon-Token` header) and **Microsoft Teams** (a Workflows webhook). Each has a test button and

the address is stored encrypted. In **Users** you create accounts and choose their role; a **monitor** only sees and gets notifications about what you mark as **Visible**. Every user, the administrator included, can change **their own password** with the “Password” button (it asks for the current and the new one, minimum 12 characters), and the administrator can reset anyone else’s. **Two-step verification (2FA)** is turned on with an authenticator app (QR) and can be turned off or reset from the user’s card.

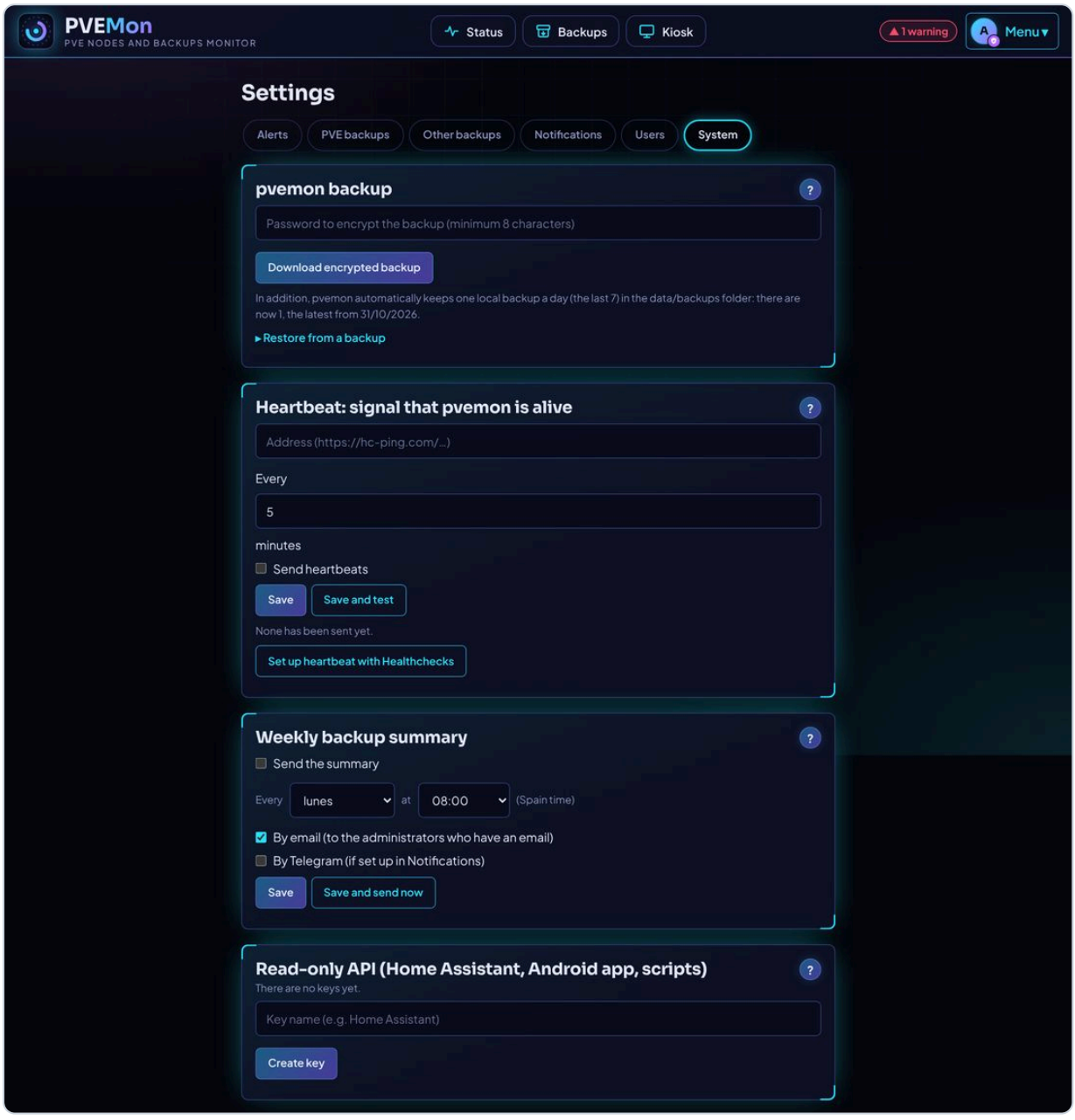


Settings: notifications (email, Telegram, webhook, Teams)



Settings: users

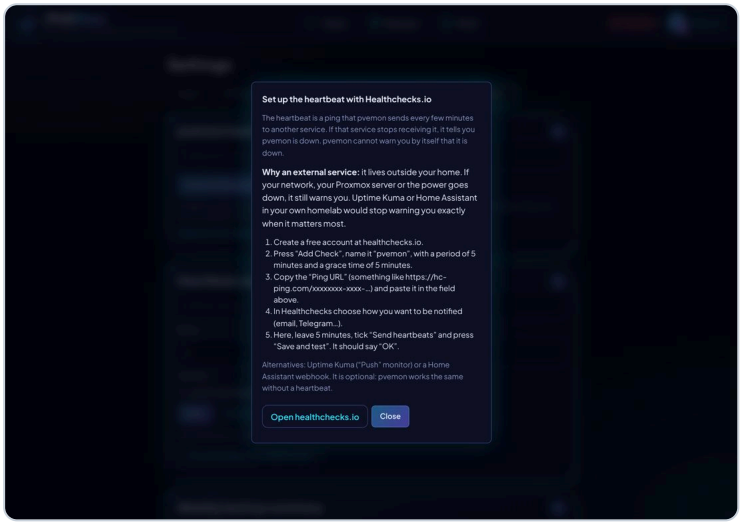
A4. System: PVEMon backup, heartbeat and weekly summary



Settings: system

- **PVEMon backup**: download a `.pvemonbak` file encrypted with a password (minimum 8 characters) and restore it from the same screen. Restoring replaces the configuration, logs everyone out and restarts PVEMon; it only works with the same secret key (`SECRET_KEY`) it was made with. PVEMon also keeps a daily local copy (the last 7).
- **External heartbeat**: PVEMon visits an address every N minutes (Healthchecks.io, Uptime Kuma, a Home Assistant webhook...) so that an outside service warns you if PVEMon stops working. The **Set up heartbeat with Healthchecks** button opens the help window.

- **Weekly backup summary**: email and/or Telegram with good and problem backups, size, time left, storage space and open alerts. Day and time are configurable, with a **Save and send now** button.
- **Backup storage space**: an alert fires when a storage stays above 90 % for 10 minutes.
- **Read-only API**: for Home Assistant or scripts. One key is created per program (shown only once; only its fingerprint stays in the database) and sent as `Authorization: Bearer pvm_…`. No key can change anything.



Help window for the Healthchecks heartbeat

A5. Security and resources

Built-in measures

- Passwords hashed with scrypt, optional **2FA** with an authenticator app and account lockout after 10 failed attempts in 15 minutes, plus a per-IP limit (5 per minute on login).
- Session cookie `HttpOnly` + `SameSite=Strict` (and `Secure` over HTTPS); the session expires after 14 days.
- Request origin check, security headers (CSP, anti-iframe, no-referrer, HSTS over HTTPS) and generic errors.
- Tokens, passwords and keys stored **encrypted** with `SECRET_KEY`; each node's certificate fingerprint is pinned on first contact.
- Protected admin routes: a monitor gets "forbidden" and only sees what is marked **Visible**.
- PVEMon only reads: Proxmox PVEAuditor token, read-only Docker proxy and, if used, SSH limited to three read commands.
- Audit log of important actions.

Recommendations

- Publish PVEMon only over HTTPS (Nginx Proxy Manager) and **close port 3000** to the rest of the network.
- Enable 2FA for all users and include `SECRET_KEY` and the `data` folder in your backups.
- Limit the Docker proxy port 2375 to PVEMon's IP.
- Install PVEMon as an app so notifications carry its icon.

Server resources

Resource	Need
Processor	1 core (almost no usage)
Memory	~180 MB idle; reserve 512 MB
Disk	Image ~400 MB plus data (tens of MB after months)
Network	In: port 3000. Out: nodes (8006), Docker (2375), email, Telegram, push, webhooks and NFS

A6. Recent changes

The full history, version by version, is in the **ChangeLog** button of the menu.

Version	What is new
1.5.x	PVEMon in Spanish and English (selector on login and in the menu, language per user, also in alerts, emails, push, Telegram and summaries); dependencies with no vulnerabilities; version ready to publish as a Docker image; 1.5.1 and 1.5.2 fix the kiosk, “My notifications” and the API backup list, and add the PDF guide
1.4.x	Uptime counter with mood faces; several kiosks and the Kiosk button menu; undo/redo and arrow keys in the panel editor
1.3.0	IP address of VMs and LXCs, guest agent state and node address on its sheet
1.2.x	Input and moved-to folders for the backup mailbox, Edit button on sources and tasks, wildcards in file checks
1.1.0	Read-only API (/api/v1) with keys; disk power cycles; automatic check of pending updates
1.0.0	First stable version: no BETA label, subtitle “PVE nodes and backups monitor”, updated dependencies and new README
0.14.x	PVEMon only reads (no updating or rebooting; read-only SSH script and “Remove access”); Disks before “Connection and maintenance”; health and all disks on the Dashboard; titles with icons; ChangeLog button and © Dagarin; two-line header on phones; no “All OK”; swipe between panels in a group; own icon in notifications; password change for the logged-in user
0.13.0	Disk monitoring (SMART, wear, temperature, ZFS), webhook, Teams, alert snoozing and 2FA
0.12.1	Kiosk: rotating groups with transition, main kiosk, text box, multi-select and custom sizes
0.11.x	Kiosk mode: canvas editor, MINI tiles, summaries, per-panel theme and preview
0.10.x	File checks on a network disk (NFS), icons per backup type and narrow-screen tweaks
0.9.x	VM and LXC menus, sheets with rings, Nodes as a list and management in the node sheet
0.8.0	Encrypted PVEMon backup, heartbeat, maintenance mode, weekly summary and backup-space alert
0.7.x	Backups screen, backup size and Status and Backups buttons in the title bar